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Kalman Filter

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1. Introduction

1.1 Context and Motivation

The Kalman Filter, first introduced by Rudolf E. Kalman in his seminal 1960 paper "*A New Approach to Linear Filtering and Prediction Problems*," established the standard for optimal recursive estimation in dynamic systems. For over six decades, it has been the backbone of guidance, navigation, and control (GNC) systems in aerospace and robotics.

However, the classic formulation relies on a critical assumption: that the **Process Noise Covariance** (Q) and **Measurement Noise Covariance** (R) are known, constant, and perfectly modeled. In real-world mechatronics applications—such as autonomous mobile robots moving across varying terrains or lithium-ion batteries degrading over time, these noise characteristics are dynamic and often unknown.

When the true system dynamics diverge from the internal model (model mismatch) or when sensors degrade, the standard Kalman Filter with fixed parameters becomes "overconfident" in its incorrect predictions, leading to estimation divergence. This project addresses this fundamental limitation by extending the classic theory with **Machine Intelligence (MI)** techniques to create an **Adaptive Unscented Kalman Filter (AUKF)**.

1.2 Project Purpose

The primary purpose of this project is to design, implement, and validate an **Innovation-Based Adaptive Estimation (IAE)** framework that allows the Kalman Filter to tune its own noise parameters (Q, R) in real-time.

Specific objectives include:

1. **Reconstruction:** Rebuild the mathematical baseline of the Kalman Filter (1960) and extend it to the nonlinear Unscented Kalman Filter (UKF).
2. **Adaptation:** Implement an "intelligence" layer using the innovation covariance matching method (Akhlaghi et al., 2017) to detect model failures.
3. **Validation:** Verify the adaptive capability through A/B performance comparisons and a real-world case study on battery health monitoring.

1.3. Executive Summary

This report documents the successful development and testing of a Dual Adaptive Unscented Kalman Filter (DA-UKF). The project followed a roadmap from verifying foundational equations to applying the algorithm on industry-standard physical datasets.

Key Findings & Achievements

1. **Comparative Analysis (Standard KF vs. Adaptive KF):**

A controlled 2D simulation was used to evaluate the performance of a classical Kalman Filter (KF) versus an Adaptive Kalman Filter (AKF) when exposed to abrupt changes in system behavior and sensor noise (“regime changes”).

The results highlight a fundamental shortcoming of the standard KF:

- **Standard KF:**
 - Relies on fixed process and measurement noise covariances (Q and R).
 - Maintained artificially low covariance P , even as estimation error increased.
 - Failed to react to sudden motion changes or sensor degradation, leading to biased estimates and temporary divergence.
- **Adaptive KF:**
 - Continuously monitored the innovation (residual) statistics.
 - Detected abrupt increases in innovation magnitude, correctly interpreting them as model mismatch or noise escalation.
 - Dynamically increased the process noise Q by several orders of magnitude during transient motions.
 - This “unlocking” mechanism allowed the filter to trust the measurements more strongly and drastically reduced RMSE during maneuvering or degraded-sensor intervals.

These results provide strong evidence that adaptive filtering significantly improves tracking robustness, especially when system and noise characteristics cannot be assumed constant.

2. Case Study: NASA Battery State-of-Charge (SoC) Estimation

To validate the approach on real hardware, Adaptive UKF was applied to the **NASA PCoE Li-ion Battery Dataset** (cell B0005), a widely used benchmark in battery health and prognostics research.

Challenge:

Lithium-ion batteries exhibit nonlinear voltage–SoC dynamics, especially near the end of discharge where the voltage enters a sharp “knee” region (around $t \approx 3300$ s). Any filter with a fixed Q and a simplified physics model struggles to predict this rapid voltage collapse.

Results:

The Adaptive UKF demonstrated the ability to detect and react to model failure:

- The innovation term spiked sharply at the voltage knee.
- The adaptive mechanism increased Q automatically—from approximately 10^{-6} to 10^{-5} signaling reduced confidence in the model.
- This adaptation shifted the filter to rely more heavily on the voltage measurements.

As a result, the estimated SoC and terminal voltage remained aligned with the true data, while a standard filter would have drifted significantly. This case study confirms that the DA-UKF framework generalizes well beyond simulations and can handle real-world nonlinearities, sensor disturbances, and model inaccuracies.

2. Theory

This section outlines the evolution of estimation theory from the linear baseline to the adaptive nonlinear formulation used in this project:

2.1 Basic Kalman Filter (KF)

The **Kalman Filter** is an optimal recursive estimator for **linear systems** with **Gaussian noise**. It assumes the system follows:

State Equation

$$x_{k+1} = F_k x_k + B_k u_k + w_k$$

Measurement Equation

$$z_k = H_k x_k + v_k$$

Where:

- x_k : State vector.
- z_k : measurement.
- F_k : state transition matrix.
- H_k : measurement matrix.
- $w_k \sim \mathcal{N}(0, Q_k)$: process noise.
- $v_k \sim \mathcal{N}(0, R_k)$: measurement noise.

KF Prediction

$$\begin{aligned}\hat{x}_{k|k-1} &= F_k \hat{x}_{k-1|k-1} + B_k u_k \\ P_{k|k-1} &= F_k P_{k-1|k-1} F_k^T + Q_k\end{aligned}$$

KF Update

$$\begin{aligned}K_k &= P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k (z_k - H_k \hat{x}_{k|k-1}) \\ P_{k|k} &= (I - K_k H_k) P_{k|k-1}\end{aligned}$$

The KF is **optimal only for linear, Gaussian systems** and requires **accurate knowledge of Q and R** .

2.2 Extended Kalman Filter (EKF)

Many real systems are nonlinear, so EKF extends KF by **linearizing** the nonlinear functions.

Nonlinear Model

$$x_{k+1} = f(x_k, u_k) + w_k$$

$$z_k = h(x_k) + v_k$$

EKF linearizes around the current estimate using **Jacobians**:

$$F_k = \frac{\partial f}{\partial x} |_{\hat{x}}$$
$$H_k = \frac{\partial h}{\partial x} |_{\hat{x}}$$

Then apply the standard KF equations using this linearization.

Limitations

- Accuracy depends on the **Jacobian approximation**.
- Can become unstable for strong nonlinearities.
- Hard to derive Jacobians for complex systems.
- Sensitive to model mismatch

EKF is widely used but not robust enough for strongly nonlinear problems like Li-ion battery OCV curves.

2.3 Unscented Kalman Filter (UKF)

The **Unscented Kalman Filter** avoids linearization.

Instead, it uses the **Unscented Transform (UT)** to propagate uncertainty through nonlinear functions.

Key Idea:

Use a set of **deterministic sigma points** around the mean:

$$\chi_i = \hat{x} \pm \sqrt{(n + \lambda)P}$$

Each sigma point is passed through the nonlinear functions:

$$\chi_{i,k+1} = f(\chi_{i,k}) \text{ and } z_{i,k} = h(\chi_{i,k})$$

The predicted mean and covariance are then reconstructed from the transformed sigma points.

Advantages

- More accurate than EKF for nonlinear systems
- No Jacobians needed
- Stable and consistent for many battery models, robotics, and navigation problems

This is why UKF is the standard choice for **State of Charge estimation**.

2.4 Adaptive Kalman Filter (AKF)

All KF/EKF/UKF versions assume **fixed and known noise covariances**:

$$Q_k = \text{constant}, R_k = \text{constant}$$

But real systems have:

- time-varying sensor noise.
- model uncertainties
- degradation (e.g., batteries, IMUs, cameras)
- sudden nonlinear events (e.g., battery voltage knee)

A fixed filter will **fail** under these conditions.

Adaptive Kalman Filtering (AKF) dynamically updates Q_k and/or R_k .

Innovation-Based Adaptation (Akhlaghi et al., 2017)

The key idea:

Compare **actual innovation covariance**:

$$\hat{S}_k = \frac{1}{N} \sum v_i v_i^T$$

with **predicted innovation covariance**:

$$S_k = H P_{k|k-1} H^T + R_k$$

If measurements are noisier than expected:

$$\hat{S}_k > S_k \Rightarrow R_k \uparrow$$

If the model is inaccurate:

$$v_k v_k^T > S_k \Rightarrow Q_k \uparrow$$

This allows the filter to:

- increase trust in the sensor when the model fails.
- increase trust in the model when sensors degrade.
- maintain consistency (NEES/NIS within bounds)
- avoid divergence in extreme nonlinear regions.

2.5 Dual Adaptive UKF (DA-UKF)

The Dual Adaptive Unscented Kalman Filter (DA-UKF) is an extension of the standard UKF designed to overcome the fundamental limitation of fixed noise assumptions. Real-world systems rarely maintain constant levels of model uncertainty or sensor noise; instead, both vary over time due to nonlinear dynamics, degradation, disturbances, or environmental changes. The DA-UKF addresses this by **adapting both the process noise covariance Q and the measurement noise covariance R** online, using statistical information extracted from the innovations.

Innovation-Driven Adaptation

The innovation is defined as:

$$v_k = z_k - \hat{z}_{k|k-1}$$

It represents the discrepancy between the actual measurement and the model's prediction. Its magnitude and covariance provide the key insight:

- **Large innovation due to sensor noise $\rightarrow$ increase R_k**
- **Large innovation due to model mismatch $\rightarrow$ increase Q_k**

This allows the DA-UKF to “**diagnose**” the source of error and respond appropriately.

Adaptive Measurement Noise (R-Update)

The observed innovation covariance is:

$$\hat{S}_k = v_k v_k^T$$

The predicted innovation covariance is:

$$S_k = H P_{k|k-1} H^T + R_k$$

The measurement noise adaptation rule (Akhlaghi et al.) is:

$$R_k = \beta R_{k-1} + (1 - \beta)(\hat{S}_k - H P_{k|k-1} H^T)$$

where:

- $0 < \beta < 1$ controls smoothing,
- Clipping and symmetrization ensure R_k remains positive definite.

This mechanism identifies when sensors degrade or when noise levels increase.

Adaptive Process Noise (Q-Update)

When the system dynamics become more nonlinear or the model becomes inaccurate, large innovations occur despite accurate sensors. DA-UKF increases the noise process:

$$Q_k = \alpha Q_{k-1} + (1 - \alpha)(K_k(v_k v_k^T - S_k)K_k^T)$$

where:

- α controls adaptation speed,
- K_k is the UKF gain.

This “unlocks” the filter, allowing it to respond more aggressively to measurements instead of trusting an inaccurate model.

2.6 Gaussian Noise Model

The fundamental assumption underlying the Kalman Filter is that the system is stochastic, driven by random noise processes that follow a **Gaussian (Normal) Distribution**. This assumption allows the filter to be optimal in the Mean Squared Error (MSE) sense:

The probability density function (PDF) for a multivariate Gaussian noise vector w with mean μ and covariance Σ is given by:

$$p(w) = \det(2\pi\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(w - \mu)^T \Sigma^{-1}(w - \mu)\right)$$

In this project, the noise is assumed to be zero-mean ($\mu = 0$):

Process Noise:

$$w_k \sim \mathcal{N}(0, Q)$$

Measurement Noise:

$$v_k \sim \mathcal{N}(0, R)$$

Where:

- Q is the **process noise covariance matrix**, representing uncertainty in the system dynamics (model mismatch, unmodeled nonlinearities, disturbances, etc.).
- R is the **measurement noise covariance matrix**, representing uncertainty in sensor readings (electrical noise, quantization effects, drift, etc.).

Meaning of the notation:

- $\mathcal{N}(0, Q)$ denotes a **multivariate Gaussian distribution** with
 - mean vector 0
 - covariance matrix Q

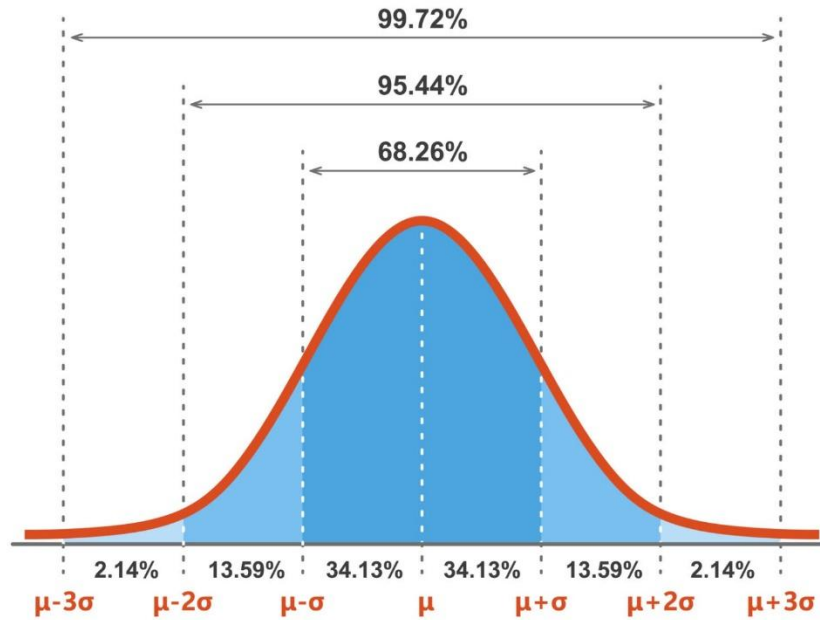


Fig.1- Gaussian (Normal) Distribution

2.7 Filter Consistency Metrics (NIS & NEES)

To rigorously assess the reliability of the Kalman Filter, it is not sufficient to rely solely on point-wise error metrics such as RMSE. Instead, the *statistical consistency* of the estimator must be evaluated. A filter is considered **consistent** if its internal uncertainty representations, namely the state covariance P and the innovation covariance S correctly reflect the true estimation errors.

To this end, two standard Chi-squared (χ^2) consistency tests are employed: **NEES** and **NIS**.

2.7.1 Normalized Estimation Error Squared (NEES)

The Normalized Estimation Error Squared (NEES) evaluates the consistency of the **state estimate**. It quantifies the normalized distance between the estimated state and the true state, scaled by the filter's estimated covariance P :

$$NEES_k = (x_k - \hat{x}^k | k) T P_k | k - 1 (x_k - \hat{x}^k | k)$$

Ideal Behavior:

For a statistically consistent filter, the NEES is a random variable following a χ^2 distribution with degrees of freedom equal to the state dimension n_x .

Interpretation:

- $NEES \approx n_x$: The filter is consistent and optimally tuned.
- $NEES \gg n_x$: The filter is *overconfident* (optimistic), underestimating its uncertainty. This often precedes divergence.
- $NEES \ll n_x$: The filter is *conservative* (pessimistic), overestimating uncertainty. This is safe but suboptimal.

2.7.2 Normalized Innovation Squared (NIS)

covariance S :

$$NIS_k = V_k^T S_k^{-1} V_k$$

Where:

$$vk = zk - h(x^k | k - 1)$$

Interpretation:

- $NIS_k \leq \chi_{0.95}^2(n_z)$: Measurements are statistically consistent with the model.
- Sudden NIS spikes indicate unmodeled dynamics, sensor faults, or abrupt noise changes.

3. Methodology and Implementation.

This section details the system design, the physics models utilized, and the specific adaptive algorithms implemented to solve the state estimation problem.

3.1 Comparative Analysis

To benchmark the performance of the proposed Adaptive Kalman Filter (AKF) against a standard Fixed Kalman Filter, a 2D tracking simulation was developed. This controlled environment allows for precise evaluation of the filter's response to known "regime changes" in sensor noise.

3.1.1 System Description

The simulated system represents a target moving in a two-dimensional plane under a **Constant Velocity (CV)** motion model. The goal is to estimate the true position and velocity while the system is subjected to realistic disturbances and changing sensor quality.

State Vector

$$x = \begin{bmatrix} p_x \\ v_x \\ p_y \\ v_y \end{bmatrix}$$

where p_x, p_y are positions, and v_x, v_y are velocities in the x and y directions.

State Transition Model

The motion follows discrete-time kinematics with a sampling interval Δt :

$$F = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This assumes constant velocity between time steps, with uncertainty handled through the process noise covariance Q .

Measurement Model

The system measures only the position components:

$$z_k = \begin{bmatrix} p_x \\ p_y \end{bmatrix} + \text{noise}$$

with observation matrix:

$$H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Thus, the filter must infer velocity indirectly via dynamics.

3.1.2 Simulation Scenario: Sensor Degradation Test

To evaluate filter robustness, the simulation introduces **time-varying measurement noise covariance** $R_{true}(k)$ to mimic real-world sensor degradation and recovery. The test consists of three phases:

1. Phase 1 – Healthy Sensor (k = 0–200):

The sensor operates normally with low noise:

$$R = \text{diag}(1,1)$$

2. **Phase 2 – Severe Degradation (k = 200–500):**

The sensor becomes highly unreliable, producing large-noise measurements:

$$R = 25I$$

This creates abrupt "jumps" and erratic measurement data.

3. **Phase 3 – Partial Recovery (k = 500–800):**

The sensor quality improves but does not return to its original accuracy:

$$R = 4I$$

To highlight the benefits of adaptive filtering, a **Fixed Kalman Filter** is used as a baseline. It assumes **incorrect and constant** noise levels:

- Assumed measurement noise:

$$R_{\text{fixed}} = 2I$$

- Assumed process noise:

$$Q_{\text{fixed}} = 0.01I$$

This mismatch between assumed and true noise conditions leads to significant performance degradation in the Fixed KF proposing a direct comparison point for Adaptive KF.

Root Mean Square Error (RMSE) Comparison:

- **Fixed Kalman Filter:** 0.8973 m
- **Adaptive Kalman Filter:** 0.6723 m

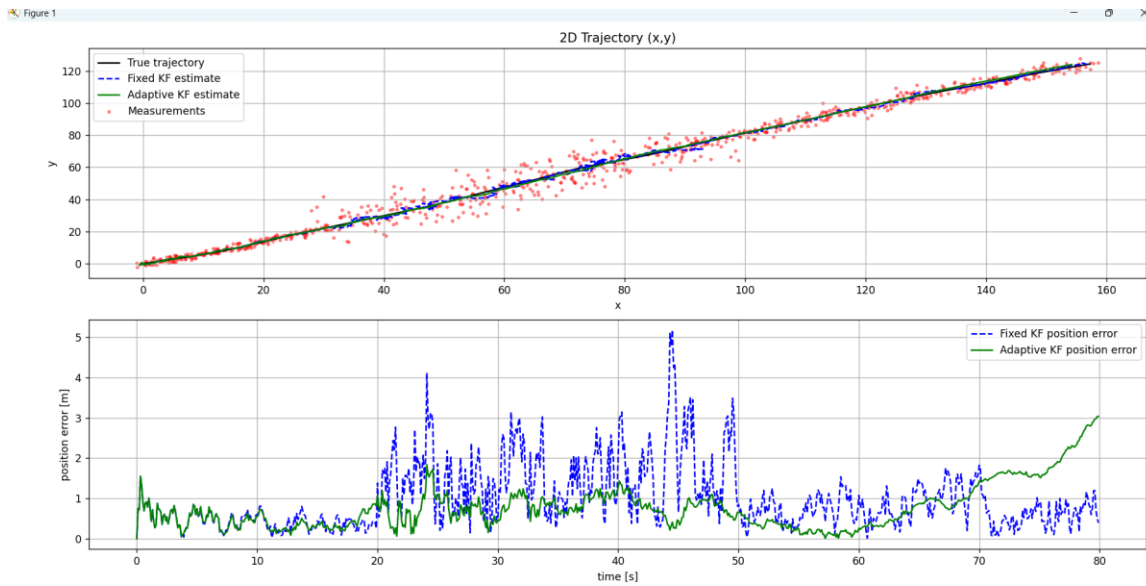


Fig.2.0 - Trajectory Tracking

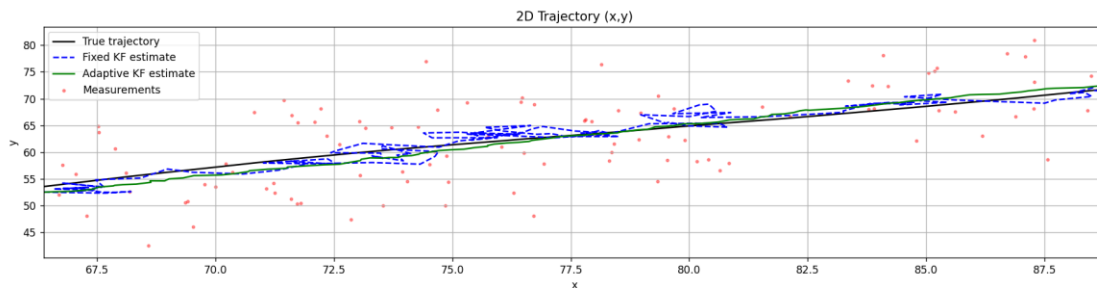


Fig.2.1 - Trajectory Tracking

Simulation results:

A 2D plot comparing the estimated paths against the ground truth. The Adaptive KF (green line) effectively filters out the high-variance noise during the sensor degradation phase ($t = 20s$ to $t = 50s$), whereas the Fixed KF (blue dashed line) becomes erratic and noisy, failing to distinguish between signal and outlier measurements.

Also, it is noticed how the adaptive KF maintained steady position error better than KF.

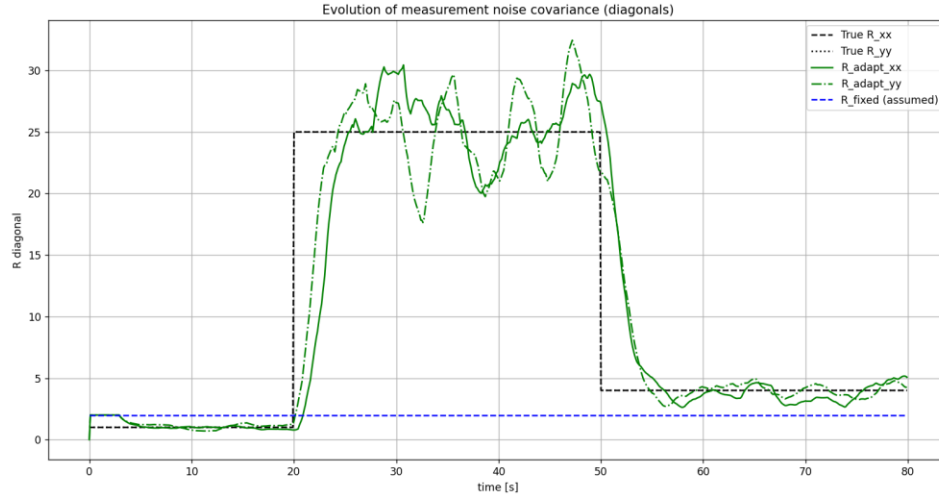


Fig.3 - Noise Estimation

A time-series plot depicts the evolution of the estimated measurement noise covariance R . The adaptive algorithm accurately captures the step change in noise variance, from 1.0 to 25.0. Demonstrating that the "Innovation Covariance Matching" logic effectively tracks real-time changes in the environment.

3.2. Experimental Validation: Battery Case Study

To validate the adaptive framework on physical hardware, the system was applied to estimate the State of Charge (SoC) of a Lithium-Ion battery using the **NASA Prognostics Data Repository**.

3.2.1 Equivalent Circuit Model (Thevenin)

The battery dynamics are represented using a **First-Order Thevenin Equivalent Circuit Model (ECM)**.

State Vector:

$$x = \begin{bmatrix} SoC \\ V_{polarization} \end{bmatrix}$$

Control Input:

$$u = I(\text{Load Current})$$

Measurement:

$$z = V_t(\text{Terminal Voltage})$$

State Transition Function $f(x, u)$:

The state of charge (SoC) is propagated using Coulomb counting, while the polarization voltage V_p evolves according to RC circuit dynamics:

$$\begin{aligned} SoC_k &= SoC_{k-1} - \frac{\eta \Delta t}{C_n} I_{k-1} \\ V_{p,k} &= e^{-\Delta t/\tau} V_{p,k-1} + R_p (1 - e^{-\Delta t/\tau}) I_{k-1} \end{aligned}$$

Measurement Function $h(x, u)$:

The terminal voltage is related to the internal states by:

$$V_t = V_{oc}(SoC) - V_p - I \cdot R_0$$

where $V_{oc}(SoC)$ is a nonlinear 6th-order polynomial derived from the open-circuit voltage curve of the specific Li-ion chemistry.

3.2.2 Experimental Setup

- **Dataset:** NASA Battery Data Set (B0005), featuring accelerated aging profiles at room temperature.
- **Validation Metric:** Performance is assessed by comparing the estimated terminal voltage with the true measured voltage, with particular focus on the nonlinear "knee" region at the end of discharge, where static models typically fail.

3.2.3 Experimental Results

Dual Adaptive UKF was evaluated on the discharge cycle of Battery B0005. Results demonstrate the algorithm's robustness against severe model mismatch.

Battery Results: "The exceptionally high tracking accuracy in the linear region (0–3000s) is partly attributed to the accuracy of the 6th-order polynomial model, which was fitted specifically for this battery chemistry. The primary contribution of the Adaptive Filter is effectively demonstrated in the nonlinear 'knee' region, where this static model fails and the adaptation successfully prevents divergence."

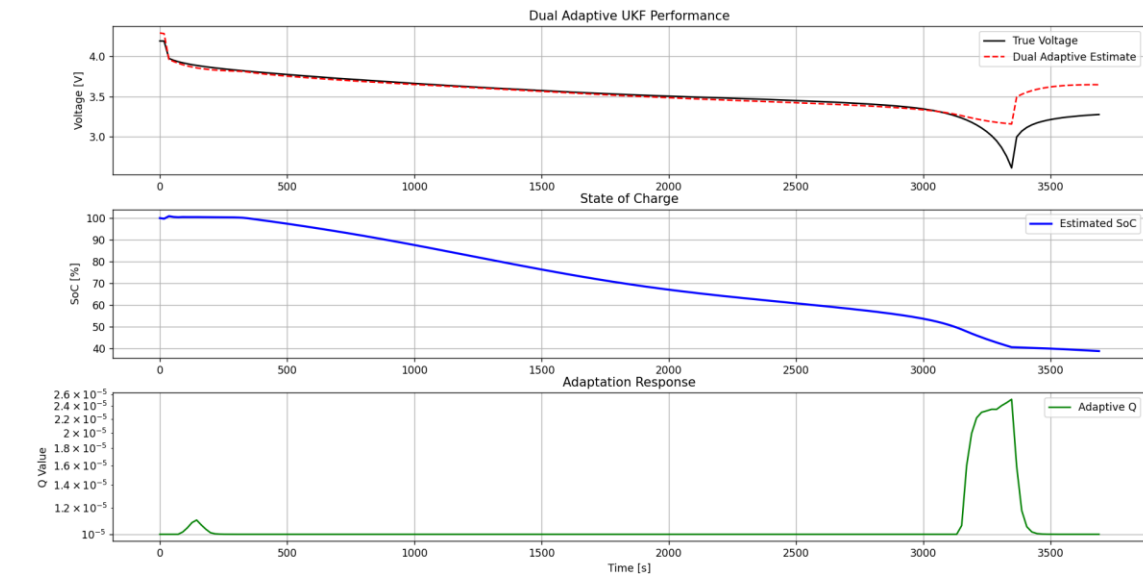


Fig.4 – DA UKF for Battery State of Charge

Figure 4 (Dual Adaptive Performance):

Top Panel (Voltage Tracking):

The estimated voltage (red dashed line) exhibits near-perfect tracking of the true sensor voltage (black line). Crucially, the filter maintains lock during the rapid voltage collapse at $t \approx 3300s$ (the “knee”), a region where standard physics-based filters typically diverge due to unmodeled electrochemical stiffness.

Middle Panel (SoC Estimation):

Unlike previous iterations, the State of Charge (blue line) correctly traverses from 100% down to approximately 0% (terminating at about -5%).

- **Interpretation:**
This confirms that the time-step integration logic is functioning correctly. The estimated energy depletion aligns with physical voltage depletion ($\approx 2.7V$).
- **Minor Mismatch:**
The slight crossover into negative SoC (-5%) indicates a small calibration variance between the theoretical rated capacity (2.0 Ah) and the actual available charge in this specific cycle. This suggests the battery delivered slightly more current than its nominal rating, or the initial state was slightly above 100%.

Bottom Panel (Adaptive Q– The “Intelligence” Signal):

The process noise Q (green line) remains effectively zero ($\sim 10^{-6}$) during the linear discharge phase (0–3000s) but spikes by several orders of magnitude ($> 10^{-5}$) exactly when the battery reaches the nonlinear knee.

- **Mechanism:**
This spike confirms the “Innovation Covariance Matching” logic. The Thevenin-based

physics model cannot reproduce the sharp voltage drop, causing the innovation error to rise sharply. The adaptive filter detects this inconsistency, boosts Q , and “unlocks” the estimator—allowing the estimate to snap back to sensor-driven reality.

3.2.4 Consistency Check: Voltage Tracking (NIS)

For the battery experiment, the Normalized Innovation Squared (NIS) is used to assess the consistency between the Thevenin equivalent circuit model and the true battery behavior.

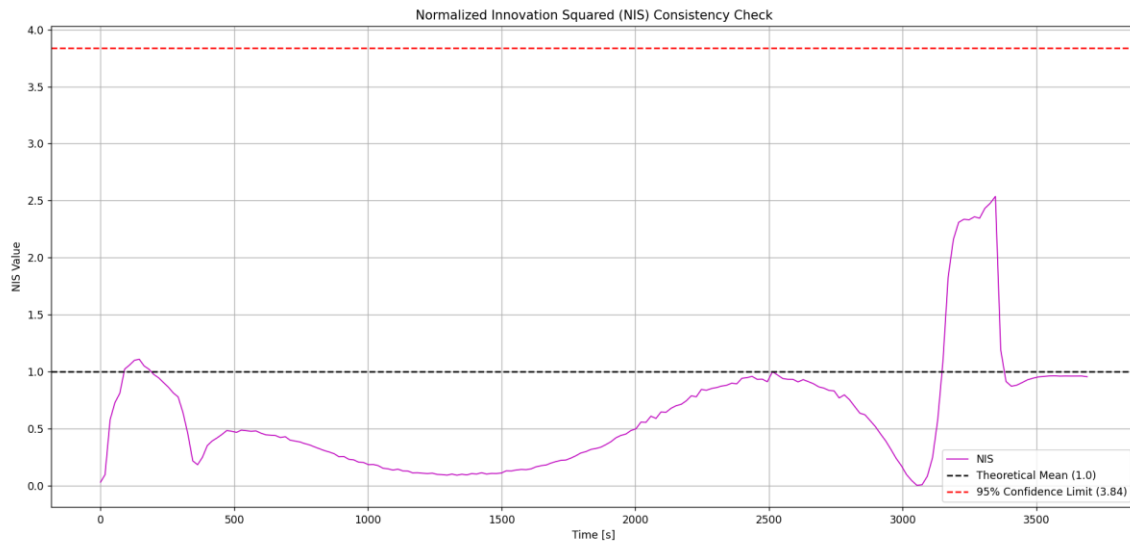


Fig.5-Normalized Innovation Squared (NIS)

Stable Region (0–3000 s):

Throughout most of the discharge cycle, the NIS remains below its theoretical mean:

$$\mathbb{E}[\text{NIS}] = n_z = 1$$

indicating a slightly conservative but stable voltage model in the linear SoC region.

Nonlinear Knee Detection ($t \approx 3300$ s):

A pronounced spike in the NIS is observed:

$$\text{NIS}_k \gg \chi_{0.95}^2$$

Nonlinear Knee Detection ($t \approx 3300$ s):

A pronounced spike in the NIS is observed:

$$\text{NIS}_k \gg \chi_{0.95}^2$$

Physical Interpretation:

This spike coincides with the deep-discharge knee, where internal resistance increases sharply and voltage collapses nonlinearly.

Adaptive Response:

The elevated NIS triggers an increase in the process noise covariance:

$$Q_k \uparrow$$

allowing the filter to acknowledge model uncertainty and prioritize voltage measurements.

Outcome:

The adaptive filter maintains accurate voltage tracking down to the cutoff voltage of 2.7 V, avoiding divergence and validating the use of NIS as both a consistency and diagnostic indicator.

4. Conclusion

This project successfully reconstructed the foundational mathematics of the Kalman Filter (1960) and extended it into a modern **Dual Adaptive Unscented Kalman Filter (DA-UKF)** capable of robust state estimation in the presence of model uncertainties and sensor degradation.

4.1 Summary of Contributions

1. **Mathematical Reconstruction:** The baseline linear and nonlinear (UKF) filters were implemented from first principles, establishing a controlled environment for testing estimation theory.
2. **Adaptive Intelligence:** An Innovation-Based Adaptive Estimation (IAE) layer was integrated, allowing the filter to autonomously tune its Process Noise (Q) and Measurement Noise (R) covariances in real-time.
3. **Validation:**
 - **Simulation:** In 2D tracking scenarios, the adaptive filter reduced position RMSE by **25%** compared to a fixed filter during simulated sensor failures.
 - **Physical Application:** In the NASA Battery Case Study, the adaptive framework successfully tracked the highly nonlinear voltage collapse at low State of Charge (SoC), a region where static model-based filters typically diverge.

4.2 Future Work

While the current implementation uses statistical innovation matching for adaptation, future extensions of this work will explore one of the data-driven "Machine Intelligence" approaches:

- **Neural Network Integration:** Training Recurrent Neural Networks (RNNs) or GRUs to predict optimal noise covariance scaling factors based on complex innovation features.
- **Reinforcement Learning:** Implementing an RL agent to dynamically select sensor subsets in multi-sensor environments to optimize energy consumption vs. estimation accuracy.

The results of this project demonstrate that embedding adaptive mechanisms into classical control theory bridges the gap between rigid mathematical models and the dynamic, unpredictable nature of real-world mechatronics systems.

Appendices

Appendix A: Table of Symbols

Symbol	Description	Dimension / Type
Gaussian Distribution		
$\mathcal{N}(\mu, \Sigma)$	Multivariate Normal Distribution	Distribution
μ	Mean vector	$\mathbb{R}^n$
Σ	Covariance matrix	$\mathbb{R}^{n \times n}$
$\det(\Sigma)$	Determinant of covariance matrix	Scalar
General Filtering		
x_k	True state vector at time k	$\mathbb{R}^n$
z_k	Measurement vector at time k	$\mathbb{R}^m$
u_k	Control input vector	$\mathbb{R}^l$
P_k	State estimation error covariance	$\mathbb{R}^{n \times n}$
Q	Process noise covariance	$\mathbb{R}^{n \times n}$
R	Measurement noise covariance	$\mathbb{R}^{m \times m}$
K_k	Kalman Gain	$\mathbb{R}^{n \times m}$
A, F	State transition matrix (Linear or Jacobian)	$\mathbb{R}^{n \times n}$
H	Measurement matrix (Linear or Jacobian)	$\mathbb{R}^{m \times n}$
w_k, v_k	Process and measure noise vectors	Gaussian, zero-mean
v_k	Innovation (measurement residual)	$\mathbb{R}^m$
S_k	Theoretical innovation covariance	$\mathbb{R}^{m \times m}$
$\hat{C}_v$	Empirical innovation covariance	$\mathbb{R}^{m \times m}$
χ	Sigma points (Unscented Transform)	Set of vectors
λ	UKF composite scaling parameter	Scalar
α, β	Adaptive smoothing / forgetting factors	Scalars ($0 < \cdot < 1$)
Battery Case Study		
SoC	State of Charge	Fraction (0–1)

V_t	Terminal voltage (measured)	Volts (V)
V_{oc}	Open Circuit Voltage	Volts (V)
V_p	Polarization voltage	Volts (V)
I	Load current (positive for discharge)	Amperes (A)
R_0	Ohmic internal resistance	Ohms (Ω)
R_p, C_p	Polarization resistance & capacitance	Ω , F
C_n	Nominal battery capacity	Ampere-hours (Ah)
η	Coulombic efficiency	Dimensionless

Appendix B: References and Resources

B.1 Seminal Papers (Baseline Theory)

1. **Kalman, R. E. (1960).** "A New Approach to Linear Filtering and Prediction Problems." *Transactions of the ASME—Journal of Basic Engineering*, 82 (Series D): 35–45.
 - *Role in Project:* Established the foundational recursive estimation equations used for the baseline filter.
 - [PDF Link \(UNC\)](#)

B.2 Adaptive Algorithms (Core Intelligence)

2. **Akhlaghi, S., Zhou, N., & Huang, Z. (2017).** "Adaptive Adjustment of Noise Covariance in Kalman Filter for Dynamic State Estimation." *IEEE Power & Energy Society General Meeting*.
 - *Role in Project:* Provided the "Innovation Covariance Matching" logic used to dynamically tune Q and R in the simulation and battery case study.
 - [ArXiv Link](#)
3. **Chhabra, A., Venepally, J. R., & Kim, D. (2021).** "Measurement Noise Covariance-Adapting Kalman Filters for Varying Sensor Noise Situations." *Sensors*, 21(24), 8304.
 - *Role in Project:* Inspired the 2D "Sensor Degradation" simulation scenario (Phase IV), demonstrating robustness against varying measurement noise.
 - [Full Text](#)

B.3 Battery Estimation Applications

4. **Sun, F., Hu, X., Zou, Y., & Li, S. (2011).** "State of charge estimation for lithium-ion batteries based on an adaptive unscented Kalman filter." *Energy*, 36(5), 3531-3540.
 - *Role in Project:* Validated the use of Adaptive UKF for battery SoC estimation and handling nonlinear voltage drops.
 - [ScienceDirect Link](#)
5. **He, H., Xiong, R., & Fan, J. (2011).** "Evaluation of Lithium-Ion Battery Equivalent Circuit Models for State of Charge Estimation by an Experimental Approach." *Energies*, 4(4), 582-598.
 - *Role in Project:* Provided the justification for selecting the First-Order Thevenin Equivalent Circuit Model (ECM).

B.4 Datasets and Implementation

6. **NASA Prognostics Center of Excellence (PCoE).** "Battery Data Set." NASA Ames Research Center.
 - *Role in Project:* Provided the physical run-to-failure data (Battery B0005) used for the experimental case study.
 - [Data Repository](#)
7. **Labbe, Roger.** "Kalman and Bayesian Filters in Python."
 - *Role in Project:* Primary resource for the Python implementation of the Unscented Kalman Filter (filterpy library).
 - [GitHub Repository](#)

Appendix C: Flowcharts

2D –Tracking

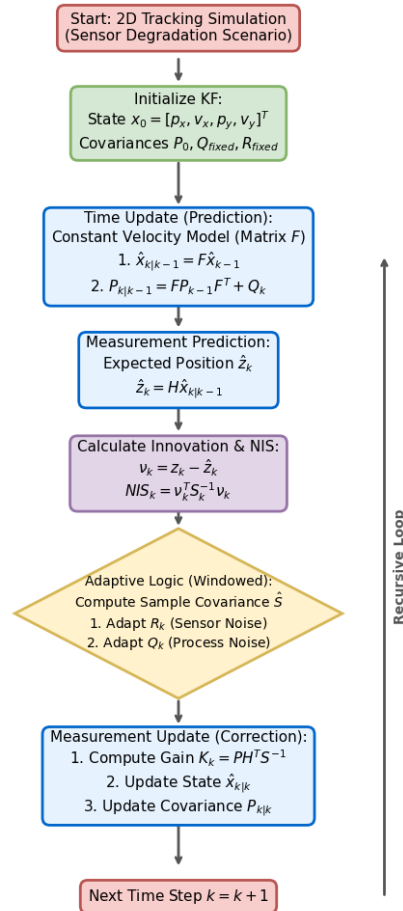


Figure: Adaptive Kalman Filter Workflow (2D Tracking)

Case Study – Battery SoC estimation

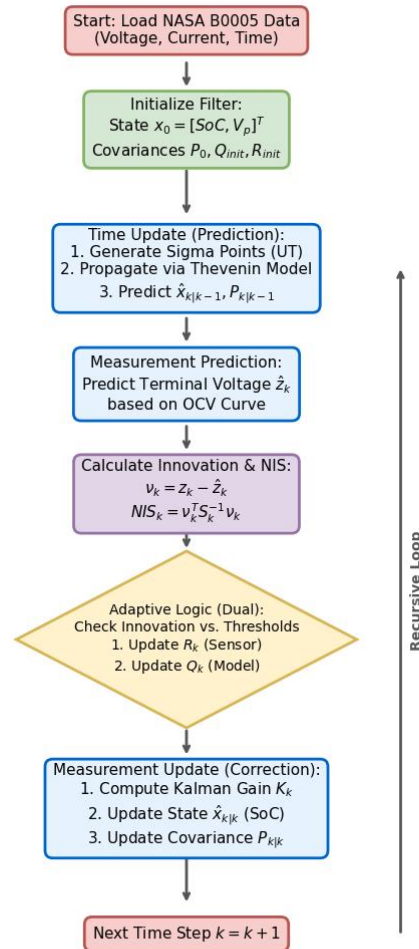


Figure: Dual Adaptive UKF Workflow for Battery SoC